

sol3_research_agent_sdk. White paper

A topic in. An evidence-backed technical white paper out.

Claude Agent SDK. Seven roles. Python owns the phases.

This is not the old config-only port. Saturday Lab 3 still fills two functions. This folder writes `paper.md`.

Read `HOW_TO_RUN.md` and `DESIGN_DOC.md`.

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Setup. Folder-local venv

```
cd solutions/sol3_research_agent_sdk
echo 'ANTHROPIC_API_KEY=sk-ant-...' >> ../../.env
task setup
```

`.venv` plus the Agent SDK. Also pins `imagen-diagrams v0.2.0` and `image-gen v2.1.0` in `.cache/`.

Do not install into Homebrew Python. Do not discover user or parent-project skills. The allowlist is those two plugins only.

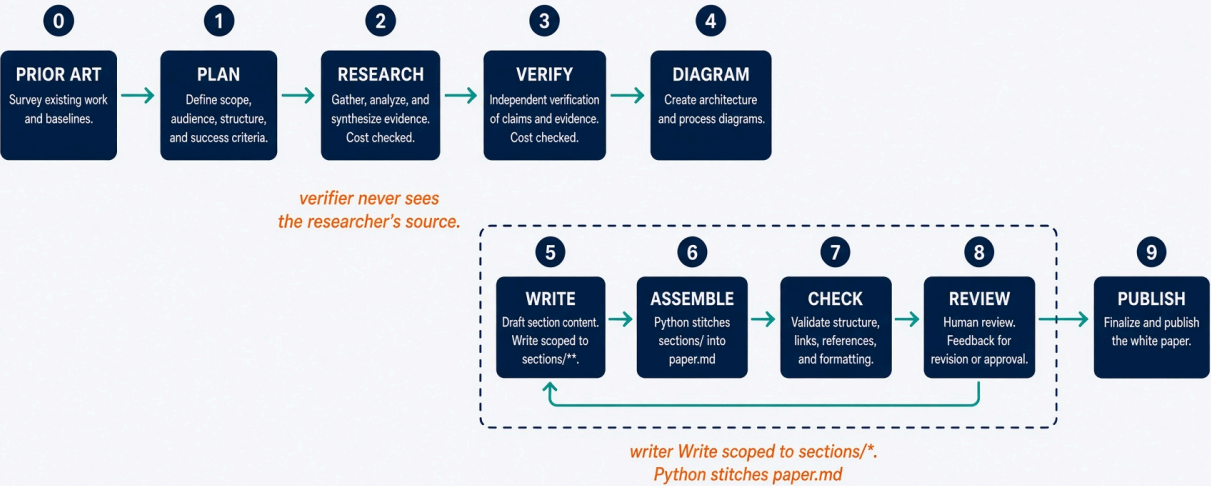
Scripts with no model

```
task table          # one writer. Everyone else prints no
task checks
task test
task demo          # recorded fixture. No key, no network.
```

If judge, researcher, or verifier prints `yes`, stop.

Starting architecture

Agent SDK white paper. Ten phases. Python owns the order.



Three exits: pass / retry / escalate. Cost checked inside research and verify, not only at the gate.

See also `docs/diagrams/architecture.svg`.

The cast. Seven roles, one writer.

orchestrator	Task	writes nothing
planner	Read, Glob, Grep	returns plan JSON. Python writes plan.json
researcher	Read + search MCP	writes nothing
verifier	Read + search MCP	writes nothing. Never sees the source.
diagrammer	Read, Glob, Grep	returns diagram source. Python renders.
writer	Read + Write	sections/** only. Cannot reach paper.md
judge	Read, Glob, Grep	writes nothing

Two places enforce scope. You need both.

`tools=[...]` decides whether a role can write at all.

One `PreToolUse` hook decides which paths. It reads `agent_type`.

A write with no `agent_type` is denied. So is a write from anyone but the writer. So is a write to `paper.md` or `claims.json`.

sol1 registered one hook per writing role. Several hooks on `Write` all run. An empty dict means no opinion. The first role that shrugs lets another role's write through.

`allowed_tools` **is not the parent's list**

It is a session-wide permission allowlist. It gates subagents too.

First live run: `allowed_tools=["Agent"]` with `dontAsk`. Every search came back denied.

The researcher did **not** answer from memory. It reported `NO RESEARCH WAS PERFORMED` and returned an empty claim list. The run escalated.

`options_for` now allows the union of what the cast holds. `Bash` is denied at the top.

MCP is declared in Python

Context7 is always configured. Perplexity is included only when `PERPLEXITY_API_KEY` is set.

`.mcp.json` holds a key, so it is gitignored. A fresh clone has no such file. Declare servers in Python or a missing file silently cannot search.

This port has no OpenAI or Bing fallback. A missing root `.mcp.json` must not change the tool boundary.

Ten phases. Python owns the order.

0 prior_art	skip if missing	-> prior-art.md
1 plan	sections and questions	-> plan.json
2 research	claims from primary sources	-> sources.json, claims.json
3 verify	independent second look	-> verdicts.json
4 diagram	source in, Python renders	-> diagrams.json
5 write	one section at a time	-> sections/*.md
6 assemble	stitch + references	-> paper.md
7 check	seven deterministic rows	-> check.json
8 review	judge on what a script cannot	-> review.json
9 publish	secret gist, on request	-> gist.json

Phases 0 to 4 run once. 5 to 8 are the retry cycle.

Cost is checked inside the phases

`--max-usd` default 5.00. Checked inside research, inside verification, and at the gate.

Checking only at the gate is a bug this port had. A twenty-four question research phase can spend the whole budget several times over before the gate ever sees it.

Running out mid-verification leaves remaining claims `unverified`. Marking them verified because the money ran out is the lie.

The check that matters most: `sourced`

A web search cannot refute a citation that was never published. Asking a model whether a reference is real gets you a confident yes.

The only thing that catches a fabricated arXiv id or DOI is looking for it in the text that was actually retrieved.

`checks.ungrounded_identifiers`.

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White-paper acceptance runs

```
task e2e-fixture  
LIVE_E2E_MAX_USD=10 task e2e-live  
REPORT_DIR=work/e2e-loop-engineering-live task pdf  
REPORT_DIR=work/e2e-loop-engineering-live task publish-report
```

Fixture uses recorded research plus real `imagen-diagrams` figures. Live needs both keys and never publishes a gist by itself.

PDF uses Arctic Fox. `publish-report` is a secret gist. The URL is the credential.

`task clean` deletes `work/`. The `.cache/` plugins stay.

Testing skill

```
.agents/skills/e2e-test-research-report/
```

Default folder is this one.

```
task setup && task test && task checks  
LIVE_E2E_MAX_USD=10 task e2e-live
```

Use the fixture lane when credentials are absent. Do not describe a fixture lane as live.

Troubleshooting

SYMPTOM	FIX
Subagent tools denied	union of the cast, plus hook
Silent empty research	<code>mcp_servers()</code> in Python
Fabricated sections	empty claims must escalate
Bill before a gate	check cost inside research and verify
Writer rewrote <code>paper.md</code>	deny anyone but writer, deny that path
SVG in the paper	renderer must exit 2, not substitute

Recap

A topic in. A paper plus a knowledge bundle out.

1. Trigger and runtime change. Exits do not.
2. `allowed_tools` gates the children.
3. MCP in Python, or a fresh clone silently cannot search.
4. Three budgets, and cost inside the phases.
5. `sourced` catches a citation a model will bless.

The grounding contract has to hold while the wiring is wrong.